

Machine Learning-Based Remaining Useful Life Predictions and Its Application on Predictive Maintenance

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Abstract— Predictive maintenance marks a significant improvement over traditional methods by preventing failures, minimizing downtime, and reducing costs, especially in aviation where it enhances safety. This paper explores the impact of window size on Remaining Useful Life (RUL) estimation using the Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset. The performance of Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) is evaluated in handling time-series data, highlighting the importance of effective windowing techniques. Various fixed window sizes (10, 20, 25, 30, 40, 50-time cycles) with consistent overlap were tested to isolate the window size as the primary variable. Results indicate that larger window sizes generally improve the performance of LSTM models by capturing essential temporal dependencies, while CNN models perform optimally with smaller windows. The paper underscores the trade-offs between window size, accuracy, and computational efficiency, offering insights into optimal configurations for reliable RUL predictions in predictive maintenance applications.

Keywords— predictive maintenance, Remaining Useful Life, LSTM, CNN, CMAPSS dataset, window size, time-series data

I. INTRODUCTION

Predictive maintenance improves traditional methods by preventing failures, minimizing downtime, and reducing costs, especially in aviation, where it also enhances safety. This paper focuses on the benefits of using Artificial Intelligence (AI) and Machine Learning (ML) for predictive maintenance, utilizing the Commercial Modular Aero-Propulsion System Simulation (CMAPSS) dataset [1]. Remaining useful life (RUL) estimation is vital for timely maintenance, predicting how long machinery will operate before needing repair [2]. Techniques include model-based (using physical models and data), data-driven (using machine learning on historical data), and hybrid (combining both approaches). Model-based techniques require deep system knowledge, while data-driven methods need substantial data and careful handling. Hybrid methods balance both for accurate predictions [3]. Challenges in RUL estimation include data quality, operational variability, modeling complexity, technological integration, and costs. Overcoming these is key to advancing predictive maintenance strategies. The CMAPSS dataset, used for this paper, contains multivariate time series data from a fleet of engines, critical for simulating and studying engine degradation and operational behaviors under various conditions.

Data serve as the foundational fuel for RUL models, where the structuring of this data critically depends on the chosen window size. Effective windowing captures essential trends and variations, directly influencing model performance and accuracy. The window, defined by its length, determines the number of consecutive data points analyzed, capturing the granularity of trends. For example, non-overlapping windows spanning 10-time cycles illustrate how data can be segmented for analysis in Figure 1. The CMAPSS dataset's structure enables detailed studies of engine components and their functions within the propulsion system, enhancing the development of predictive maintenance models.

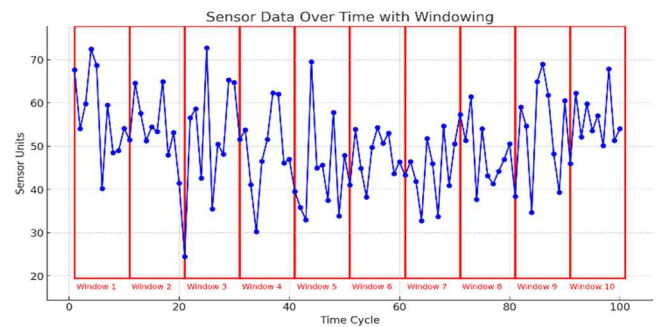


Figure 1: Windowing Techniques for Sensor Data Over Time

Overlap between windows, expressed as a percentage of the window size, ensures no significant patterns are missed and maintains continuity in data analysis. Figure 2 shows sensor data with 25% overlapping windows, depicted by alternating red and blue rectangles.

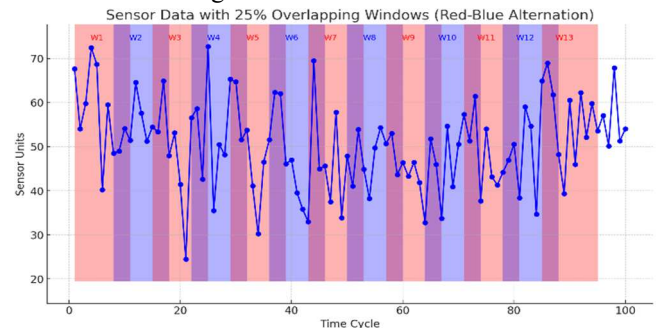


Figure 2: Overlapping Techniques Using 25% for Sensor Data Over Time

II. RELATED WORK

The authors of the paper [4] explore the efficacy of integrating regression outputs derived from varying window

lengths to enhance the classification of pedestrian maneuvers. This investigation specifically focuses on assessing the potential benefits of utilizing regression results across different window sizes. The regression methodology employed involves a rolling window technique, which outlines the regression process.

The paper highlights the impact of window size variability on the accuracy of the regression models. Shorter window sizes are noted for their ability to detect quick changes in data, although they are more prone to noise and measurement errors, leading to unstable predictions. In contrast, longer window sizes yield more consistent predictions but tend to be slower in adapting to actual changes in the dataset, potentially causing delays in predictive accuracy. These observations underscore the trade-offs between window sizes in the context of pedestrian maneuver classification [4]. Figure 3 shows the windowing technique.

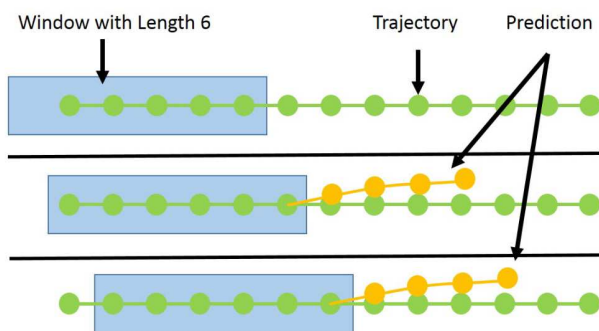


Figure 3: Regression process with rolling window [4].

Sateesh Babu et al. [5] explore a deep CNN-based regression approach was developed to estimate the RUL of machinery components. The CNN model was evaluated across two public datasets, including the CMAPSS dataset, and compared against traditional regression models like Multi-layer Perceptron (MLP), Support Vector Regression (SVR), and Relevance Vector Regression (RVR). Figure 4 illustrates the architecture proposed by Sateesh Babu et al. [5]

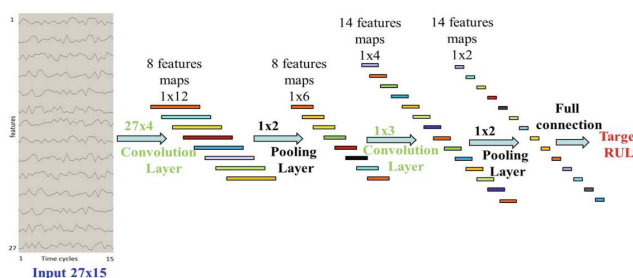


Figure 4: The proposed CNN architecture for estimating RUL [5].

The Root Mean Square Error (RMSE) results for the CNN approach consistently outperformed the other models on all CMAPSS sub-datasets, which supports the effectiveness of CNNs in capturing more informative features from time series data for RUL estimation. The specific RMSE values observed for each sub-dataset under the CMAPSS data set are as follows [5]:

FD001: CNN achieved an RMSE of 18.4480

FD002: CNN achieved an RMSE of 30.2944

FD003: CNN achieved an RMSE of 19.8174

FD004: CNN achieved an RMSE of 29.1568

These results demonstrate the superior performance of the CNN model compared to traditional approaches in the domain of RUL estimation for aircraft engines, highlighting its robustness across various operating conditions and fault scenarios within the dataset.

Zheng et al. [6] explore the effectiveness of LSTM networks in predicting the RUL of components using sensor data. This paper differentiates itself by demonstrating the LSTM's superior capability to manage long-term dependencies in sequence data, which is crucial for accurate RUL estimation under varying window sizes.

The paper tested the LSTM model against traditional approaches and CNNs across three Prognostics and Health Management datasets. The LSTM models consistently outperformed other methods, achieving lower RMSE values, indicative of their higher accuracy. Notable RMSE improvements demonstrated by the LSTM approach include:

FD001: RMSE reduced to 16.14 from CNN's 18.45,

FD002: RMSE reduced to 24.49 from CNN's 30.29,

FD003: RMSE reduced to 16.18 from CNN's 19.82,

FD004: RMSE reduced to 28.17 from CNN's 29.16.

A comprehensive review of the literature was conducted to determine the most frequently used machine learning models in the domain of RUL prediction [7]. This review helped to identify which models have been most effective and popular among researchers and practitioners. The findings from this review are summarized in the Table 1, which lists the models along with the count of publications supporting their use:

Table 1: Diversity of approaches taken to model RUL [4].

Model	Count
LSTM	28
CNN+LSTM	10
CNN	10
ESN, GRU, RNN	7
Tree-based	6
Other ANN	5
Other	5
Similarity-based	4
MLP, ELM	4
Particle Filter	2

III. METHODOLOGY

This paper employs advanced machine learning techniques to predict the RUL of machinery, focusing on model selection, data preprocessing, and performance evaluation. The primary models used are LSTM networks and CNNs, selected for their effectiveness in handling time-series data and feature extraction. Data preprocessing involves signal smoothing and normalization to ensure quality and relevance. Sensor selection is based on variance, correlation with RUL, and impact on model accuracy, balanced with computational efficiency considerations. The dataset is structured by

segmenting it into overlapping windows, with padding applied where necessary, to capture the sequential nature of the data. Performance metrics such as, R-squared (R^2) and Mean Relative Error (MRE) are used to evaluate the models. Hyperparameter tuning is performed using Random Search, known for its efficiency in high-dimensional spaces, and K-fold cross-validation is employed to ensure robustness and

generalizability. This systematic approach allows for a comprehensive evaluation of the impact of window size on model performance, providing insights into optimal configurations for accurate and reliable RUL predictions in predictive maintenance application. Figure 5 shows the methodology stages.

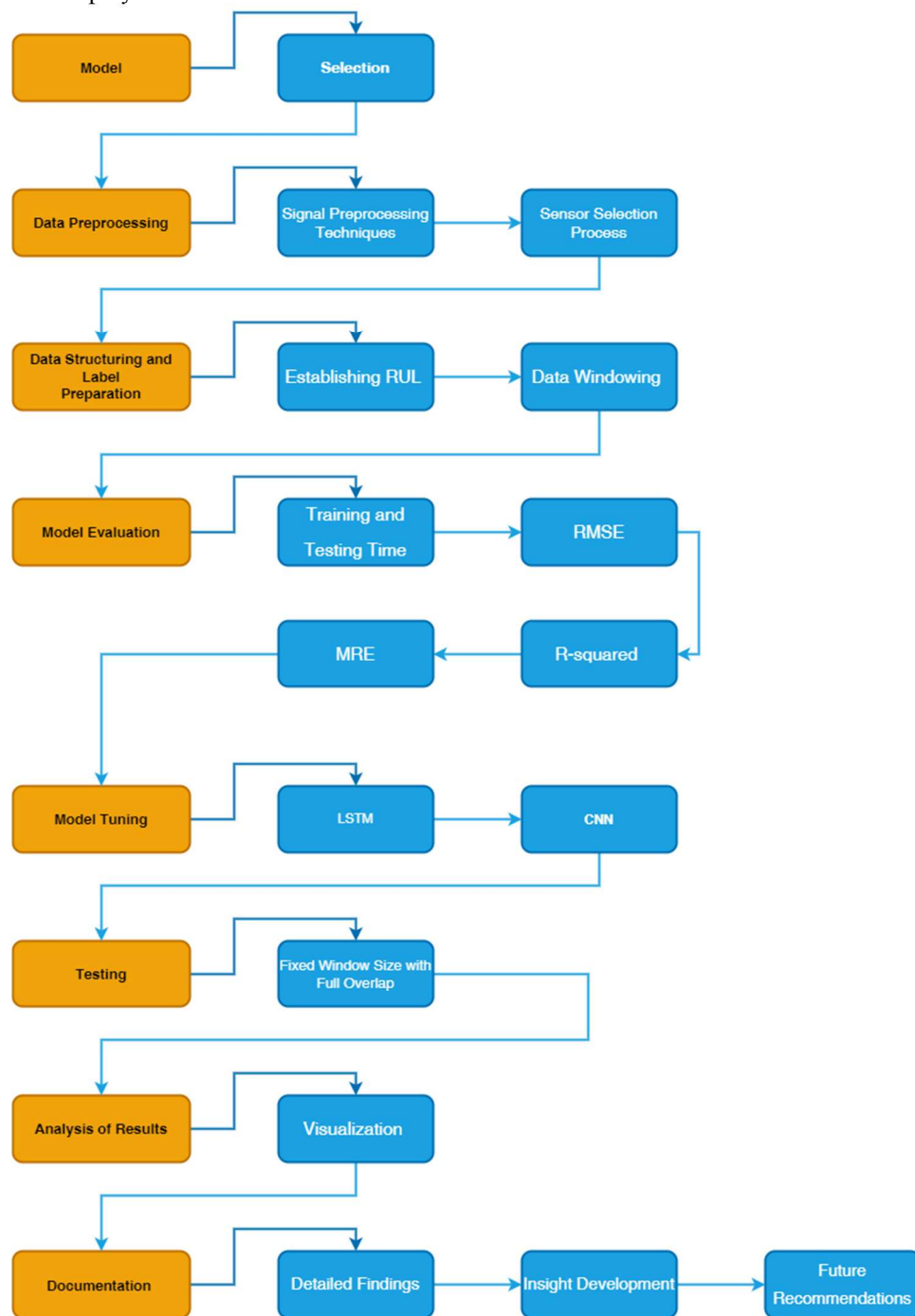


Figure 5: Methodology Stages.

IV. TESTING AND RESULTS

This section evaluates the impact of various fixed window sizes on the predictive accuracy of the model. The aim is to determine how different dimensions of the input window

influence model performance, with a consistent overlap across tests to isolate window size as the primary variable.

Multiple Fixed window sizes of 10, 20, 25, 30, 40, and 50 cycles were tested, each with full overlap. This approach was designed to examine the correlation between window size and accuracy in model predictions for both LSTM and CNN

models. Table 2 contains the summary of the best performance of the test 1 results.

Table 2: Test 1 results -The best model and performance for each dataset.

Dataset	Window Size	Test RMSE	Test R2	Best Model
FD001	50	13.77	0.89	LSTM
FD002	40	24.25	0.80	LSTM
FD003	25	14.93	0.87	CNN
FD004	30	25.44	0.78	LSTM

A. RMSE evaluation metrics.

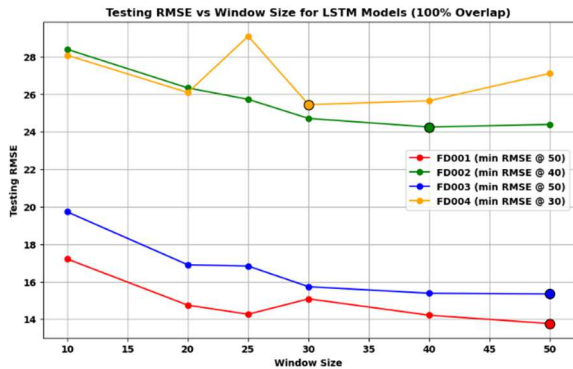


Figure 6: LSTM - Testing RMSE Vs. Window Size

LSTM Observation: For FD001 and FD003, the largest window size correlates with the lowest RMSE value, indicating better prediction accuracy with more extensive data per window. In contrast, for FD002 and FD004, while larger window sizes also lead to lower RMSE values, the optimal window size is not necessarily the largest, differing from the pattern observed in FD001 and FD003.

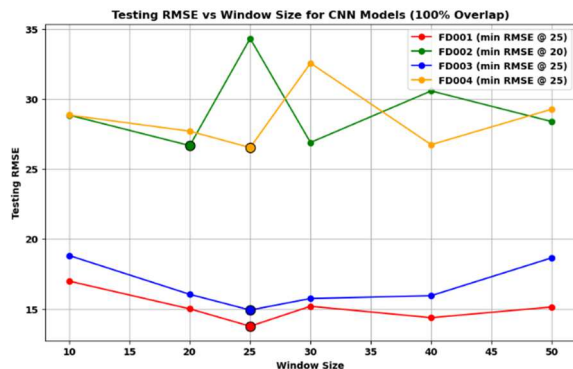


Figure 7: CNN - Testing RMSE Vs. Window Size

CNN Observation: Generally, for all datasets, a window size of 25 correlates with the lowest RMSE values, suggesting optimal prediction performance at this size. However, FD002 deviates from this pattern, where a window size of 20 yields the lowest RMSE, indicating a different optimal window configuration for this dataset.

B. R^2 evaluation metrics.



Figure 8: LSTM - Testing R2 Vs. Window Size.

LSTM Observation: For FD001 and FD003, higher window sizes correspond to increased R2 values, signaling enhanced model performance with larger amounts of data per window. This trend complements the reduced RMSE observed with greater window sizes. Conversely, for FD002 and FD004, although R2 values improve with expanding window sizes, the peak R2 values do not always occur at the maximum window sizes. This indicates that the best window size for optimizing model performance in these datasets may not be the largest, suggesting a more intricate data structure or diverse data quality.

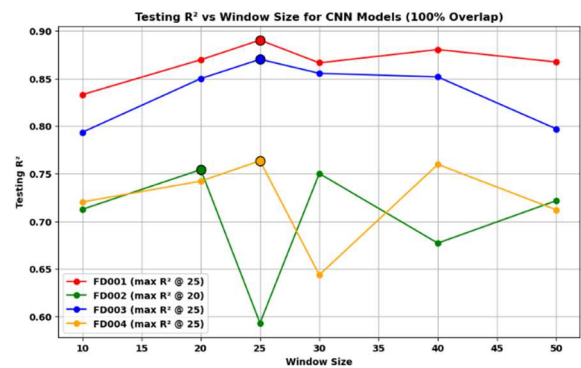


Figure 9: CNN - Testing R^2 Vs. Window Size

CNN Observation: Similarly, for most datasets, a window size of 25 is associated with higher R2 values, indicating that this size efficiently captures more of the variance in the data, which aligns with the reduced RMSE at this window size. For FD002, however, the peak R2 value occurs at a window size of 20, mirroring the RMSE findings and suggesting that the optimal window size for both minimizing error and maximizing explained variance may differ for this dataset.

C. MRE Evaluation metrics.

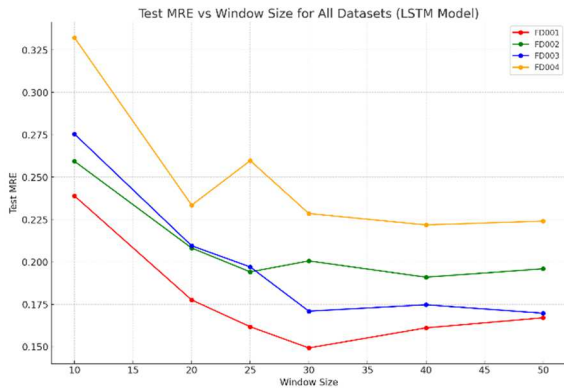


Figure 10: LSTM- Testing MRE Vs. Window Size.

LSTM Observation: For the LSTM model, larger window sizes generally improve performance for FD001 and FD003, as indicated by the decreasing test MRE. FD002 and FD004 show improvement up to certain window sizes (around 25 for FD002 and 20 for FD004), beyond which additional data per window does not significantly enhance performance. This suggests that while larger windows help initially, there's a point beyond which gains are minimal.

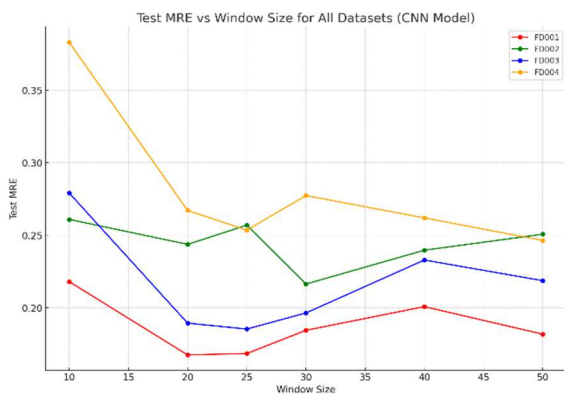


Figure 11: CNN- Testing MRE Vs. Window Size.

CNN Observation: For the CNN model, an optimal window size of around 20-25 is observed for most datasets. FD001 and FD002 achieve minimal test MRE around window size 25, while FD003 and FD004 show optimal performance around window size 20, indicating that this range captures the necessary data variance and minimizes error effectively.

D. Computational training and testing time.

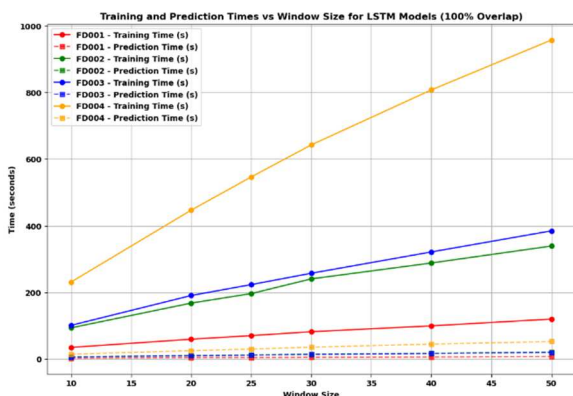


Figure 12: LSTM - Testing and Training Time Needed Vs. Window Size.

LSTM Observation: It is evident that the computation time required is in direct proportion to the window size across all datasets; as the window size increases, both training and prediction times increase. Generally, FD004 exhibits longer computation times compared to the other datasets, highlighting differences in dataset complexities.

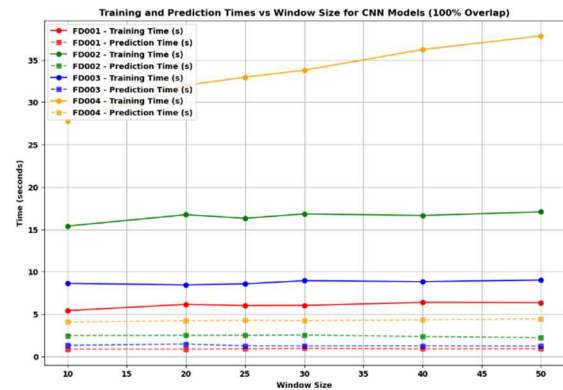


Figure 13: CNN - Testing and Training Time Needed Vs. Window Size

CNN Observation: The computational time needed shows a slight increase with larger window sizes; however, this increase is relatively modest and can be considered nearly constant across different window sizes. Generally, FD004 exhibits longer computation times compared to the other datasets, which underscores differences in dataset complexities.

V. DISCUSSION

The general observations indicate that LSTMs take longer to train than CNNs, particularly with larger window sizes, due to their sequential data processing. Additionally, LSTMs also have longer prediction times, which increase with window size, while CNNs maintain consistent prediction times.

In terms of accuracy, both models perform better with larger window sizes, but LSTMs typically achieve higher accuracy metrics, reflected in lower RMSE, MRE, and higher R^2 values.

Regarding the impact of window size, larger windows enhance LSTM performance by capturing temporal dependencies, with optimal sizes identified as 50, 40, and 30 for different datasets. CNNs, on the other hand, perform best with smaller windows, such as 25, in less complex datasets. However, the trade-offs of using larger window sizes include improved accuracy but increased computational load, necessitating a balance between accuracy and efficiency.

There are also limitations and challenges to consider. The C-MAPSS dataset, which consists of synthetic data, may not fully capture real-world engine conditions. The limited size and scope of the dataset could also restrict the generalizability of the findings. Furthermore, random search may not always identify the optimal parameters, as it depends heavily on the chosen range and distribution of values.

VI. CONCLUSION

The findings indicate that the accuracy of RUL predictions significantly varies with changes in window size across different machine learning models. LSTM models demonstrated improved prediction accuracy with larger window sizes, which allowed for capturing essential temporal dependencies more effectively. In contrast, CNN models showed optimal performance at smaller window sizes, suggesting a lesser dependency on extended historical data. This variance underscores the need for model-specific tuning of window sizes to enhance prediction accuracy.

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